Department of Electronics and Communication Engineering

Khulna University of Engineering & Technology

<u>Proposal of Electronic Circuits Design Laboratory (ECE 2200)</u>

Project Title: Design and Implementation of an Adjustable Regulated DC Power Supply with Overvoltage, Undervoltage, and Short-Circuit Protection.

Group No: B-6

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1. Title:

Design and Implementation of an Adjustable Regulated DC Power Supply with Overvoltage, Undervoltage, and Short-Circuit Protection.

2. Introduction:

A regulated DC power supply is one of the most essential instruments in electronics laboratories, research facilities, and industrial applications. Most electronic devices require a stable and controllable DC voltage for proper operation. However, fluctuations in input voltage, excessive output voltage, low voltage conditions, and short circuits can damage sensitive electronic equipment. Therefore, protection mechanisms must be incorporated into the power supply system to ensure safe and reliable operation.

This project presents the design and implementation of an adjustable regulated DC power supply with integrated Over Voltage Protection (OVP), Under Voltage Protection (UVP), and Short Circuit Protection (SCP). The system converts 220V AC mains voltage into a low-voltage DC supply using a step-down transformer, bridge rectifier, and filter capacitor. An LM317 adjustable voltage regulator is employed to provide a variable DC output. Protection circuits based on LM393 voltage comparators continuously monitor the output voltage and compare it with preset reference levels. When an abnormal condition such as overvoltage, undervoltage, or short circuit occurs, the comparator activates a transistor-relay control circuit that disconnects the load, preventing damage to connected equipment.

The proposed system provides adjustable output voltage, improved reliability, enhanced safety, and efficient fault protection, making it suitable for laboratory power supply applications and electronic testing environments.

3. Objectives:

i. To design an adjustable regulated DC power supply using LM317.
ii. To convert 220V AC mains voltage into smooth regulated DC voltage.
iii. To provide a variable output voltage suitable for different electronic circuits.
iv. To implement Over Voltage Protection (OVP) for load safety.
v. To implement Under Voltage Protection (UVP) for stable operation.
vi. To implement Short Circuit Protection (SCP) to prevent excessive current flow.
vii. To develop a relay-based isolation system for fault conditions.
viii. To improve the reliability and safety of electronic power supplies.
ix. To analyze the performance of the protection circuits under different operating conditions.

4. Proposed System Design:

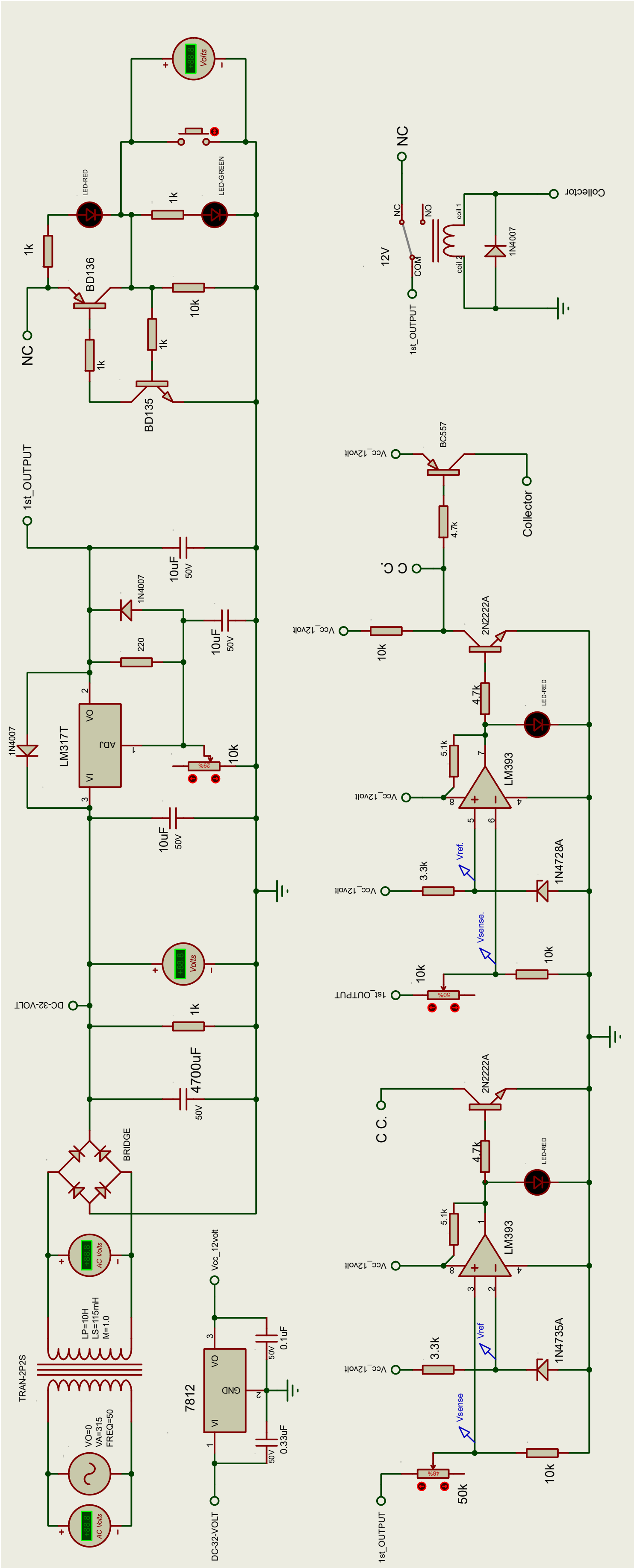

Fig: circuit diagram for Adjustable Regulated DC Power Supply

5. Proposed Budget for Implementing the Project:

Sl.	Item	Rating	Justification	Price (BDT)
1.	Transformer (2P2S)	220V AC / 24V AC, 2A	Steps down mains voltage to a safe low AC voltage for rectification and regulation.	600
2.	Capacitors	4700µF/50V, 10µF/50V, 0.33µF/50V, 0.1µF/50V	Used for ripple filtering, regulator stability, and noise suppression.	110
3.	Resistors	1kΩ, 3.3kΩ, 4.7kΩ, 10kΩ, etc.	Used for voltage division, current limiting, biasing, and LED protection.	20
4.	Variable Potentiometers	10kΩ, 50kΩ	Used for output voltage adjustment and OVP/UVP threshold setting.	75
5.	Diode (1N4007)	1000V, 1A	Used in bridge rectification and relay flyback protection.	35
6.	Zener Diode (1N4735A, 1N4728A)	6.2V, 1W, 3.3V, 1W	Provides stable reference voltage for comparator-based protection circuits and voltage monitoring and protection thresholds.	10
7.	IC 7812	12V Fixed Regulator, 1A	Supplies regulated 12V for comparators, relay, and control circuitry.	20
8.	LM393	Dual Voltage Comparator	Detects overvoltage and undervoltage conditions.	35
9.	Relay (SPDT)	12V coil	Disconnects load during fault conditions for protection.	45
10.	LM317T	Adjustable Regulator, 1.5A	Provides adjustable regulated DC output voltage.	35
11.	Transistors(NPN)	BD135, 2N2222A	Used in output indication/control stage and Drives protection and relay control circuitry.	25
12.	Transistors(PNP)	BD136, BC557	Used in complementary switching and indication circuit and for relay switching and control logic.	10
Total: 1020 BDT **In words:** One Thousand and Twenty Taka Only				

6. Work Plan (Week wise plan):

1st Week	Study the concept of adjustable power supplies and protection systems (OVP, UVP, SCP), select components, and prepare the circuit design.
2nd Week	Design and simulate the complete circuit in Proteus, verify voltage regulation and protection operation under different fault conditions.
3rd Week	Assemble the hardware on breadboard/PCB, test individual modules (rectifier, regulator, OVP, UVP, SCP, relay circuit).
4th Week	Integrate the complete system, perform final testing, troubleshoot issues, collect results, and prepare the project report and presentation.

7. Expected Outcomes:

The proposed adjustable regulated DC power supply is expected to provide a stable and controllable DC output voltage suitable for laboratory and electronic testing applications. The LM317-based regulator is expected to maintain output voltage regulation despite variations in input voltage and load conditions.

The Over Voltage Protection (OVP) circuit is expected to disconnect the load whenever the output voltage exceeds the preset safe limit, thereby protecting connected devices from damage. Similarly, the Under Voltage Protection (UVP) circuit is expected to detect abnormally low output voltage conditions and isolate the load to ensure reliable operation. The Short Circuit Protection (SCP) mechanism is expected to rapidly respond to fault conditions and prevent excessive current flow through the system.

The relay-based protection system is expected to provide complete electrical isolation during fault conditions, while LED indicators are expected to provide clear visual status information regarding normal and fault operations. Overall, the system is expected to offer improved safety, reliability, voltage regulation, and fault tolerance compared to a conventional adjustable power supply.